

1 (a) gravitational force provides/is the centripetal force

B1

$$GMm/r^2 = mv^2/r \quad \text{or} \quad GMm/r^2 = mr\omega^2$$
$$\text{and } v = 2\pi r/T \quad \text{or} \quad \omega = 2\pi/T$$

M1

with algebra to $T^2 = 4\pi^2 r^3 / GM$

A1 [3]

or

acceleration due to gravity is the centripetal acceleration

(B1)

$$GM/r^2 = v^2/r \quad \text{or} \quad GM/r^2 = r\omega^2$$
$$\text{and } v = 2\pi r/T \quad \text{or} \quad \omega = 2\pi/T$$

(M1)

with algebra to $T^2 = 4\pi^2 r^3 / GM$

(A1)

(b) (i) equatorial orbit/orbits (directly) above the equator

B1

from west to east

B1 [2]

(ii) $(24 \times 3600)^2 = 4\pi^2 r^3 / (6.67 \times 10^{-11} \times 6.0 \times 10^{24})$

C1

$$r^3 = 7.57 \times 10^{22}$$

$$r = 4.2 \times 10^7 \text{ m}$$

A1 [2]

(c) $(T/24)^2 = \{(2.64 \times 10^7)/(4.23 \times 10^7)\}^3$
 $= 0.243$

B1

$T = 12 \text{ hours}$

A1 [2]

or

$$k (= T^2/r^3) = 24^2/(4.23 \times 10^7)^3$$
$$= 7.61 \times 10^{-21}$$

(B1)

$$T^2 (= kr^3) = 7.61 \times 10^{-21} \times (2.64 \times 10^7)^3$$
$$= 140$$

$T = 12 \text{ hours}$

(A1)